

Responsibility and AI

5ARC0

2026-09-15 Lecture

Four main notions of responsibility

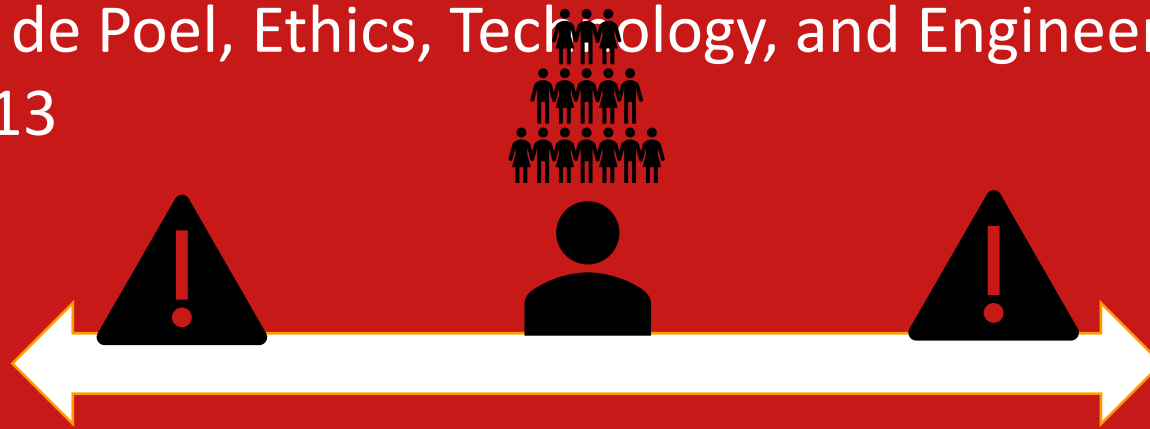
Passive (backward-looking) moral responsibility

Active (forward-looking) moral responsibility

Collective moral responsibility

Legal responsibility

Royakkers & van de Poel, Ethics, Technology, and Engineering: An Introduction, p. 13



Passive/ backward-looking moral responsibility (blameworthiness)

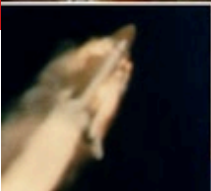
Causal Contribution: A causal connection is present between a person's action and certain consequences;

Freedom of Action: The person acted freely/ not under compulsion;

Foreseeability: The person could have reasonably known that these consequences would arise; and,

Wrong-doing: Bringing about these consequences is wrong.





**Traditional focus of
responsibility analysis:
division of
responsibility between
management and
engineer**

Image: Wikipedia

Combining responsibility and risk perspectives

People, institutions, and companies are responsible for risk (passive sense) when...

They causally contribute (increase the probability of the bad outcome);

They act freely in so doing;

They are or should be aware of the increased hazard; and,

The resulting risk is not acceptable.



Problem case for discussion: technology-created incentives

How Uber Uses Psychological Tricks to Push Its Drivers' Buttons

BY NOAM SCHEIBER

The start-up has undertaken an extraordinary experiment in behavioral science to subtly entice an independent work force to maximize company revenue.

These incentives cause safety problems (because drivers travel through residential areas near airports and drivers do without sleep). Is the designer responsible for these problems?

Definition of many hands problem

A situation in which a collective may reasonably be held responsible for an outcome, but no individual may reasonably be held responsible for that outcome.

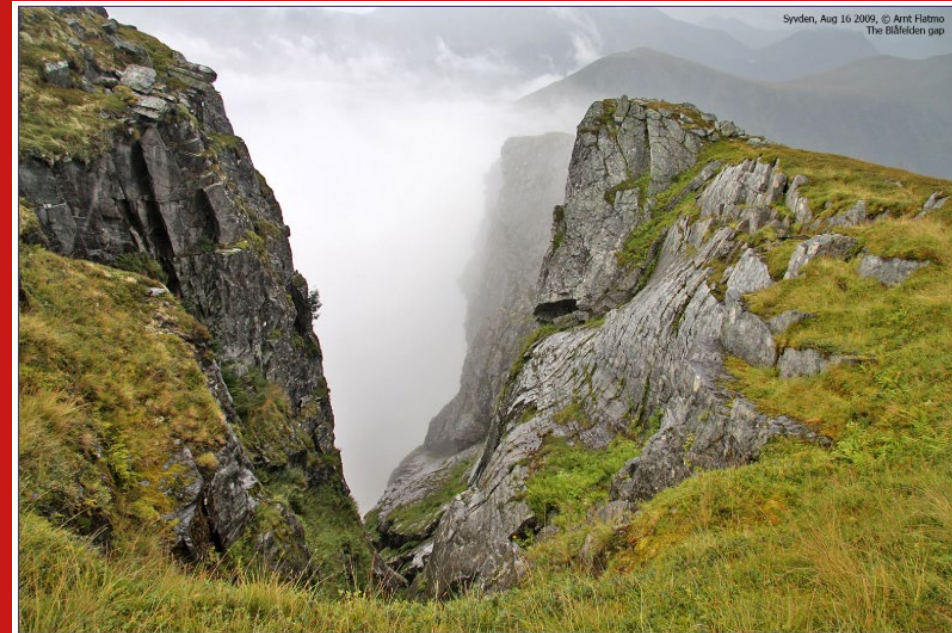
The many hands problem results from applying the criteria for moral or legal responsibility to collectives and individuals.



Responsibility gaps

The many hands problem is an example of a “responsibility gap”, where (a) it seems that we ought to hold somebody responsible for something, but (b) we cannot reasonably hold any particular entity responsible.

Responsibility gaps can be accidental or intentional (purposeful avoidance of responsibility). For example, a fashion brand might use many different suppliers with many different and changing subcontractors, in order to avoid being associated with exploitative labor conditions.



Responses to the many hands problem (and other responsibility gaps)

The many hands problem is sometimes used to motivate the importance of a clearer assignment of *active responsibility*.

Technology can be used to help designate active responsibilities.

The law can be used to help designate active responsibilities.

Active moral responsibility (role responsibility)

We can *designate* responsibility by giving a person a particular role that requires tasks. This is “active” in that it anticipates who is responsible *in the future*.

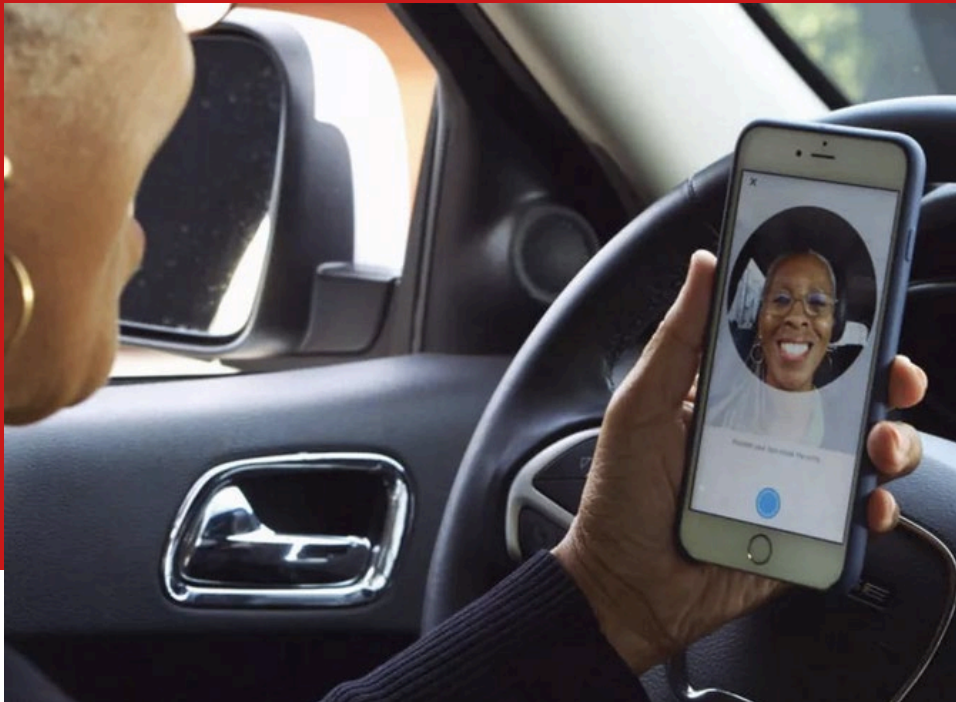


Image: Badger Herald

Using technology to partially automate role responsibility

Andrew J. Hawkins, “Uber now requires drivers to take selfies for added security; critics say some drivers have never undergone background checks,” *The Verge*, 23 September 2016.

<https://www.theverge.com/2016/9/23/13030682/uber-driver-selfie-facial-scan-fraud-security>



clearly indicate whether your chosen actor should take responsibility, and under what conditions: when others (fail to) take action, and when different outcomes materialize

Distributed role responsibilities regarding passenger safety

Driver (Professional User)	Platform designer (Software engineer)	Management (Enterprise)	Government (Society)
Follow procedures Report incidents	Provide tools for drivers Anticipate technical and security gaps	Certify drivers and vehicles for competence and security Maintain records of incidents	Require certification Enforcement

See <https://eng.uber.com>, Searching “safety”

Values in the distribution of (active) responsibility

Feasibility (Is it realistically possible to live up to the responsibilities assigned?)

Transparency to outsiders (Can people outside the organization know when to hold somebody responsible?)

Fairness (Does the distribution of tasks, and praise/blame, give people what they deserve and is it distributed equitably?)

(Royakkers & van de Poel, Ethics, Technology, and Engineering: An Introduction, p. 13 p. 266)

Allocating active responsibility

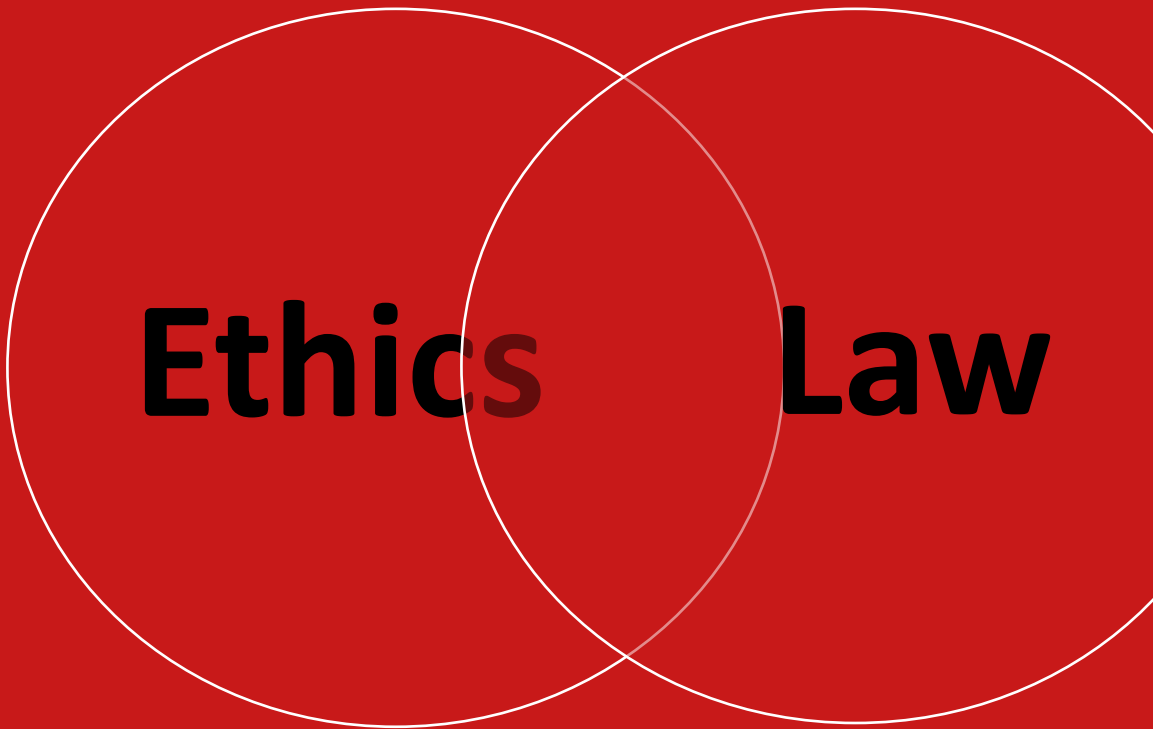
Active/ forward-looking responsibility is a matter of being assigned responsibility (or taking responsibility) for an area of activity, looking forward to the future.

This means that the responsible party puts herself in a position where she will:

- Have the ability to act
- Know what norms apply
- Adequately perceive violations of those norms
- Consider the consequences

(Royakkers & van de Poel, Ethics, Technology, and Engineering: An Introduction, p. 13)

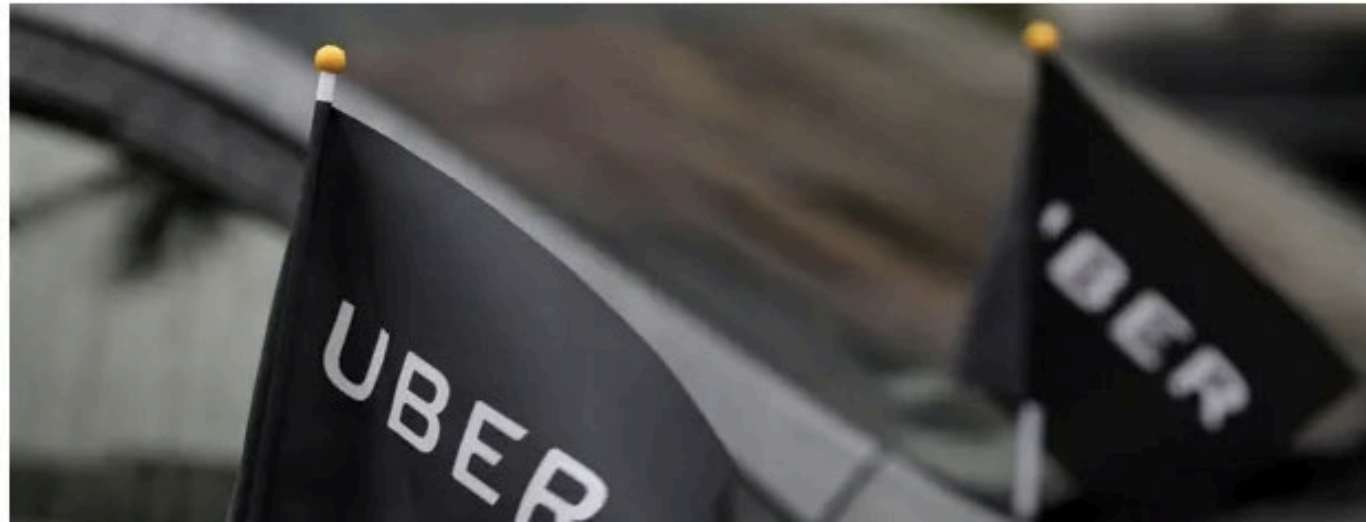
Moral responsibility vs legal



Problem case: Greyball

Greyball: how Uber used secret software to dodge the law

Ride-hailing company hit by revelations it used custom-built tool to deceive law enforcement, while latest departure of senior staffer deepens troubles



Julia Carrie Wong, "Greyball: How Uber used secret software to dodge the law," *The Guardian*, 4 March 2017

Legal responsibility

Type 1. **Criminal responsibility**

Type 2. **Liability** is backward-looking legal responsibility in which the responsible party must pay a fine, arrange repair, or pay damages.

There are different standards possible for liability: negligence, and strict product liability



By Patrick Denker - originally posted to Flickr as IMG_5078, CC BY 2.0, <https://commons.wikimedia.org/wiki/index.php?curid=5580253>

Uber self-driving car crash (bicyclist killed)

Uber Technologies Inc was held not to be *criminally* responsible for the crash. The back-up driver, Rafaela Vasquez, could have faced charges of vehicular manslaughter. (Vasquez was watching streaming TV in the car at the time of the crash, and did not react in time to take over and avoid hitting the bicyclist.) Conceivably, Uber could have been sued for liability (wrongful death). But it was unlikely to be convicted of a crime in this case.



Legal Responsibility

Liability:

- Backward-looking responsibility according to the law
- Relates to obligation to pay a fine or repair or repay damages.

Legal and moral backward-looking responsibility overlap but can come apart in two directions.

Moral responsibility

Based on the four criteria

Informal

Connected to blame, forgiveness,
moral emotions

Legal liability

Based on legal conditions

Formal (court of law with proof
standards)

Connected to fines and damages

Negligence: the usual condition for legal liability

Proof consists of showing:

- A duty of care
- A breach of the duty
- Injury or damage
- A causal connection between the breach and the injury or damage

Foreseeability (a part of the breach of duty) is understood in terms of the reasonable person standard: a reasonable person in the position of the defendant could have foreseen the damage.

Other standards of liability

Strict liability:

- The defendant engaged in a risky activity
- There was injury or damage
- This activity caused the injury or damage

Product liability:

- A kind of strict liability in which manufacturers are liable for defects in a product, without the need to show negligence.

Relationship between notions of responsibility

Backward looking/ passive moral responsibility is basic, and has to do with linking an agent with a past event in a certain way.

Forward looking/ active moral responsibility is about *taking or being given* (backward-looking) moral responsibility in regard to future events.

Legal responsibility concerns punishment and compensation as part of a system meant to ensure justice. It has additional requirements such as provable harm, efficacy, and feasibility.

Collective responsibility refers to ways of assigning responsibility when there are multiple actors and automated decision making, responding to the so-called *many hands problem*.

Cases for discussion

ChatGPT/ 2023 hallucination of court cases

From reading: CO deployment of a killer robot that kills an innocent civilian



Complications: Responsibility for actions vs omissions

Actions: Intentional behaviors with a causal impact originating with (the motion of) an agent and attributable to them.

Omissions: Non-actions or failures to act (i.e., no motion), attributable to an agent.

People can be responsible for (the foreseeable effects of) both actions and omissions.



Automation and responsibility

automation

full automation

human in the loop

human must
override system in
some conditions

system must
override human in
some conditions

<https://www.youtube.com/watch?v=f0P1lkzy8To>
<https://www.cnet.com/pictures/the-increasingly-autonomous-robots-of-war-pictures/>



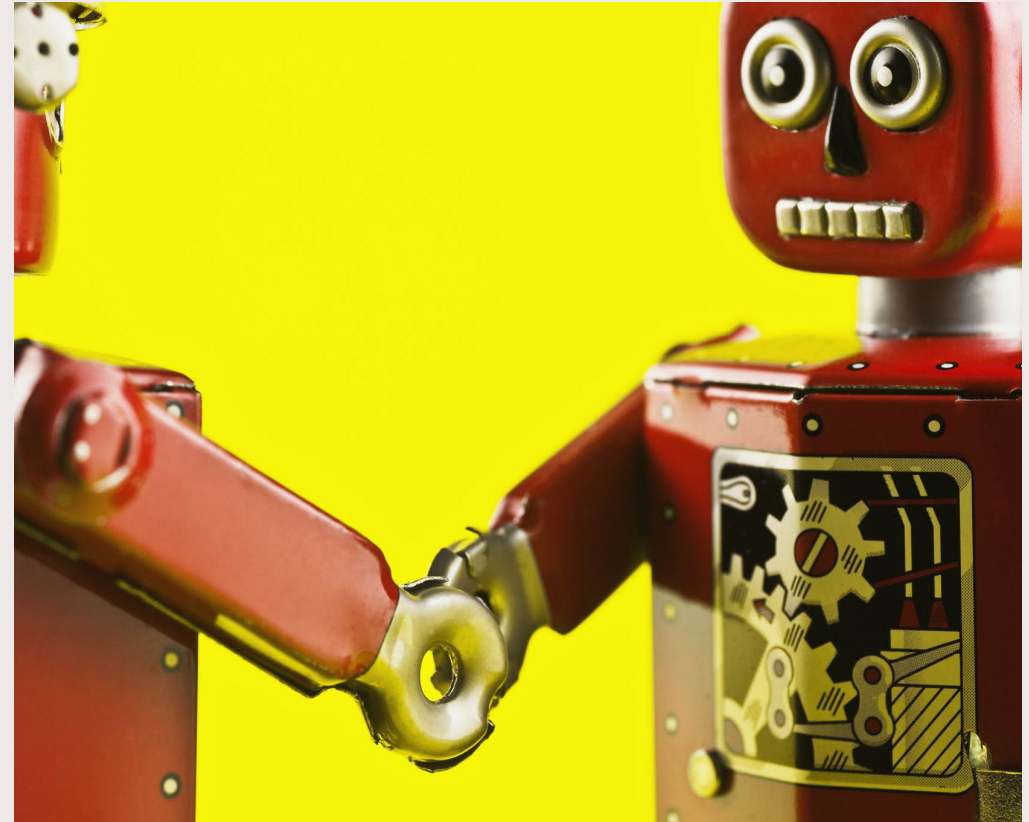
Can a machine be responsible?

Responsibility contains condition that machines do not satisfy.

In addition, there is a “retribution gap” (Danaher 2016): it does not make sense to blame or punish a machine.

One reason this might matter is that a machine cannot feel guilt, suffer, or correct its own behavior in response to blame.

See CHLEP Ch. 5 for a similar argument.



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Responsibility of primary and secondary operators



<https://www.drivingschoolsupplies.ie/store/p1/dual-controls-for-driving-schools.html>

Responsibility and design

“many of the choices made by designers can be seen as decisions about what should be delegated to a machine and what should be left to the initiative of human actors.”

Akrich 1992, P. 216

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Image:
<https://industrialscripts.com/stage-directions/>

Solutions to the problem of responsibility and failure conditions: assigning forward-looking responsibility more strictly

Can we write scripts more clearly and broadly to anticipate more failure conditions?

Can we insulate the automated system from potential failure conditions?

Can we enforce the scripts?



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Scripts

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Are there sometimes responsibility gaps due to AI?

What would be an argument for or against this position? See CHLEP Ch. 5.

Wrapping up

Types of responsibility

Four conditions on passive responsibility

Responsibility gaps

Delegation of tasks via automation